# Question ID: df9c5a1d

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Reading and Writing | Information and Ideas | Command of Evidence | Hard |

**Question**

Juvenile Plants Found Growing on Bare Ground and in Patches of Vegetation for Five Species

| Species | Bare ground | Patches of vegetation | Total | Percent found in patches of vegetation |
|---|---|---|---|---|
| ***T. moroderi*** | 9 | 13 | 22 | 59.1% |
| ***T. libanitis*** | 83 | 120 | 203 | 59.1% |
| ***H. syriacim*** | 95 | 106 | 201 | 52.7% |
| ***H. squamatum*** | 218 | 321 | 539 | 59.6% |
| ***H. stoechas*** | 11 | 12 | 23 | 52.2% |

Alicia Montesinos-Navarro, Isabelle Storer, and Rocío Perez-Barrales recently examined several plots within a diverse plant community in southeast Spain. The researchers calculated that if individual plants were randomly distributed on this particular landscape, only about 15% would be with other plants in patches of vegetation. They counted the number of juvenile plants of five species growing in patches of vegetation and the number growing alone on bare ground and compared those numbers to what would be expected if the plants were randomly distributed. Based on these results, they claim that plants of these species that grow in close proximity to other plants gain an advantage at an early developmental stage.

Which choice best describes data from the table that support the researchers' claim?

**Answer**

A. For all five species, less than 75% of juvenile plants were growing in patches of vegetation.

B. The species with the greatest number of juvenile plants growing in patches of vegetation was *H. stoechas*.

C. For *T. libanitis* and *T. moroderi*, the percentage of juvenile plants growing in patches of vegetation was less than what would be expected if plants were randomly distributed.

D. For each species, the percentage of juvenile plants growing in patches of vegetation was substantially higher than what would be expected if plants were randomly distributed.

**Correct Answer: D**

**Rationale**

Choice D is the best answer because it provides the most direct support from the table for the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. The table shows the total number of juvenile plants from five species that were found growing on bare ground and in patches of vegetation as well as the percentage of the total number of each species that were growing in patches of vegetation. For each of the five species, more than 50% of the juvenile plants were growing in patches of vegetation. The text notes, however, that a random distribution of plants across the landscape should result in only about 15% of the plants being found in patches of vegetation. In other words, for each of the five species, the percentage of juvenile plants found growing in patches of vegetation was substantially higher than could be explained by chance alone. This finding supports the claim in the text: if plants growing in patches are overrepresented among plants that have survived to the juvenile stage, as the data show they are, then it suggests that it's advantageous for plants at an early stage of development to grow in patches of vegetation.